

Louis Ledoux

Philosophiae Doctor

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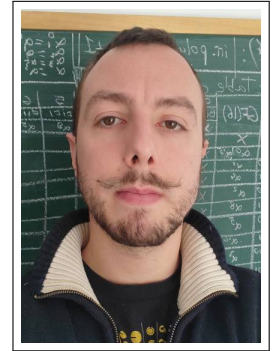
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“Post Hoc Ergo Propter Hoc”

Research Interests

- **Computer Architecture:** Floating-Point Units, Systolic Arrays, Matrix-Matrix Multiply (MMM) Units, GPUs, FPGAs, ASIC Design
- **Computer Arithmetic:** Number Representations, FloPoCo, Posit & IEEE 754 Standards, Fixed-Point Polynomial Approximations, AccuracyEnergy Tradeoffs
- **High Performance Computing:** BLAS, GEMMs, Heterogeneous Workloads, Numerical Analysis, Supercomputing
- **End-to-End Design Philosophy:** From mathematical problem to silicon implementation, integrating MLIR-based compilation and open-source silicon flows for transparent and reproducible design

Experience

2025 – Now **Postdoctoral Researcher**, INSA/INRIA Emeraude, Lyon

Working on arithmetic compilation and optimization through the integration of *MLIR* and *FloPoCo*. The research focuses on bridging high-level mathematical reasoning with low-level hardware synthesis.

- Developing multi-level arithmetic transformations from real-valued expressions to fixed-point polynomial approximations for ASIC and FPGA targets.
- Extending MLIR with custom dialects to support arithmetic abstraction, reasoning, and approximation (RealArith and FixedPointArith).
- Automating the generation of hardware architectures directly from DSP-oriented code (e.g., Faust), enabling end-to-end lowering from mathematical expressions to silicon.
- Presented results at *DSD 2025*, *EuroLLVM 2025*, and *EuroLLVM 2026*.

2018 – 2024 **Researcher**, Barcelona Supercomputing Center (BSC) - RoMoL/CAOS/SONAR, Barcelona

During the completion of my PhD, I also worked as a researcher, publishing peer-reviewed papers and attending international conferences. The conducted research is summarized as follows:

- Explored co-designed hardware/software acceleration of posit arithmetic for FPGA and ASIC technologies, and Power9 host.
- Developed Kulisch/Quire accumulators for any floating-point representation.
- Conducted accuracy and energy budgeting tailored to workload numerical requirements.
- Designed Systolic Array architecture for HPC workloads with three directions of data flow, infinite number representations, and tailored internal precision.
- Proposed very slow and small division units targeting the SIMD/Vector paradigm at the architecture level, further improving the parallelism/latency/energy ratio.

2017 - 2018 **Hardware Engineer**, b⟨⟩com, Rennes

Coop student under a professionalization contract during my final year of study. Engaged in one year of R&D focused on FPGA acceleration in the cloud, aiming to evaluate the feasibility of these novel solutions.

- Successfully integrated an IP (real-time SDR to HDR) to convert video from NVMe to NVMe through cloud FPGA.
- Developed the IP integration using HDLs and tweaked PCI-e drivers (EDMA and XDMA).
- Scheduled user processes asynchronously with OpenCL to overlap reads and writes, successfully saturating PCIe bandwidth (≈ 15.8 GB/s).
- Traveled to Xilinx XDF Frankfurt 2018 to conduct operational monitoring and analysis.

Keywords: SDAccel, OpenCL, Xilinx FPGA (Ultrascale VU9P), VHDL, SystemVerilog, AWS F1 instances, Linux driver, Linux kernel, PCIe, C++1x, C, DMA, AaaS (Acceleration as a Service), FaaS (FPGA as a Service), Virtual Machine, Docker, PCI-e virtual functions.

- July 2017 **Back End Developer**, *WaryMe*, Rennes
 Summer internship focused on developing the entire back end of a people security application.
 ○ Designed and implemented backend services and APIs.
 ○ Ensured secure data transmission using HTTPS and Let's Encrypt.
 ○ Deployed and managed the application on AWS with PM2 and Nginx.
 ○ Automated deployment processes with Jenkins.
 ○ Collaborated with front-end developers using Angular and TypeScript.
Keywords: Node.js, Git, C++, C, Angular, TypeScript, HTTPS, Let's Encrypt, PM2, Nginx, AWS, Jenkins.
- July 2016 **Back End Developer**, *ASKIA*, Paris 10ème
 During this summer internship, I developed an automated CLI tool for publishing surveys on popular platforms such as GitHub and Zendesk. **Key Concepts:** Node.js, HTTP(S), REST API, Git, Test-Driven Development (TDD), Event-Driven/Asynchronous Programming, Jasmine Framework, Mocks, Stubs.
- July 2014 **Electronics Technician**, *Radio Electronique Rennaise (R.E.R)*, Rennes
 As an electronics technician, I was responsible for repairing various electronic devices from customers, with an emphasis on audio equipment. Tasks included soldering, reverse engineering amplifier circuits, welcoming customers, and sorting/ordering electronic parts.

Education

- 2018 – 2024 **PhD Student**, *Universitat Politècnica de Catalunya (UPC)*, Barcelona
 Supervisor: Marc Casas Guix. Thesis title: “Floating-Point Arithmetic Paradigms for High-Performance Computing: Software Algorithms and Hardware Designs”. Jury members attributed the **excellent** grade.
- 2015 – 2018 **Engineering School**, *ESIR*, Université de Rennes
 Completed a three-year program culminating in an **Engineer diploma** certified by the CTI (Comité des Titres d'Ingénieurs) and a **Master's degree** (Magister) in Computer Science.
- 2013 – 2015 **Classe Préparatoire**, *CUPGE ESIR*, Université de Rennes
 Intensive program with a strong emphasis on Mathematics and Computer Sciences.

Teaching (*Total teaching load: 100h*)

- 2026 **Computer Architecture (ARC)**, *INSA Lyon*, Lyon
 Practical sessions in computer architecture for the Telecommunications curriculum, covering instruction execution, pipelining, cache memories, performance evaluation, and RISC-V assembly programming.
 Level: 1st year of engineering school (Telecommunications track, 3TC); Teaching load: 24h
- 2026 **Operating Systems (SYS)**, *INSA Lyon*, Lyon
 Core operating system mechanisms, including kernel and system calls, process scheduling, virtual memory, concurrency, and file systems.
 Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 16h
- 2026 **Long-Term Project in Compilation (PLD-COMP)**, *INSA Lyon*, Lyon
 Supervision of long-term group projects on compiler design and implementation, covering intermediate representations, program analysis, and backend code generation.
 Level: 2nd year of engineering school / Master 1 (Computer Science, INSA 4IF); Teaching load: 36h
- 2025 – 2026 **Computer Architecture (AO)**, *INSA Lyon*, Lyon
 Problem-solving and practical sessions on processor organization and instruction set architectures, based on the MSP430 microcontroller, including instruction execution, addressing modes, and memory hierarchy.
 Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h
- 2025 – 2026 **Digital Design (AC)**, *INSA Lyon*, Lyon
 Exercises and guided sessions on digital circuit design, including combinational and sequential logic, finite state machines, and introductory hardware description languages.
 Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h

International Peer-reviewed Journals

- [Bar+26] B. Barbe, **L. Ledoux**, A. Volkova, and F. de Dinechin, Reconfigurable constant multipliers: hardware models, optimization algorithm and applications, *Microprocessors and Microsystems*, 2026. • DOI:10.1016/j.micpro.2026.105270 • HAL:05510936.
- [Pop+25] M. Popoff, R. Michon, T. Risset, P. Cochard, **L. Ledoux**, et al., Frugality and circuit design for digital audio signal processing, *Revue Francophone d'Informatique et Musique*, special issue Frugalité, pérennité et création, vol. 11, 2025. • DOI:10.56698/rfim.961 • HAL:05489376.
- [Led+25b] **L. Ledoux**, P. Cochard, and F. de Dinechin, Towards optimized arithmetic circuits with MLIR, *WiPiEC Journal* (Works in Progress in Embedded Computing), vol. 11, no. 1, 2025, Salerno, Italy. Associated conference presentation at the Euromicro Conference on Digital System Design (DSD). • DOI:10.64552/wi-piec.v11i1.90 • HAL:04277512.

International Peer-reviewed Conference Papers

- [Led25] **L. Ledoux**, Design-space exploration of serialized floating-point division for DLP architectures, in *28th Euromicro Conference Series on Digital System Design (DSD 2025)*, Salerno, Italy. • DOI:10.1109/DSD67783.2025.00020 • HAL:05385247.
- [LC23a] **L. Ledoux** and M. Casas, An open-source framework for efficient numerically-tailored computations, in *33rd International Conference on Field-Programmable Logic and Applications (FPL 2023)*, Gothenburg, Sweden. • DOI:10.1109/FPL60245.2023.00011 • arXiv:2406.02579 • HAL:04277512.
- [LC22] **L. Ledoux** and M. Casas, A generator of numerically-tailored and high-throughput accelerators for batched GEMMs, in *30th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2022)*, May 2022, pp. 1–10, New York, USA. • DOI:10.1109/FCCM53951.2022.9786164 • HAL:04103774.

Tutorials, Workshops & Demos

- MLIRW26 **7th MLIR Workshop @ EuroLLVM 2026**, Dublin, Ireland, April 13, 2026. *Talk*. Progressive Arithmetic Lowering from Tensor Kernels to Synthesizable Datapaths. • HAL:05594483.
- ARITH26 **ARITH 2026**, Fulda, Germany, June 28 – July 1, 2026. *Demo*. End-to-end compilation from Python matrix multiplication and activation functions to silicon, with OpenROAD EDA catching optimization opportunities in between.
- MLIR26 **MLIR Summer School**, Galicia, Spain, August 10, 2026. *Teacher*. CIRCT and hardware generation with MLIR.

Contributions to Publications

- [Ven24] Contributor (not a listed author) to: M. Venn, Tiny Tapeout: a shared silicon tapeout platform accessible to everyone, *IEEE Solid-State Circuits Magazine*, 2024. Initiated manuscript drafting, ensured academic formatting, and contributed to revision. • URL: ieeexplore.ieee.org/document/10584359 • Evidence: github.com/mattvenn/tt-ieee-paper.

Poster Presentations

- [Led+26c] **L. Ledoux**, P. Cochard, and F. de Dinechin, “Progressive Arithmetic Optimizations: Tensor and DSP Kernels to Synthesizable Datapaths,” • *3rd FPGA Developers’ Forum (FDF) meeting*, CERN, May 27–29, 2026. • HAL:05631193v1.
- [Led+26b] **L. Ledoux**, P. Cochard, and F. de Dinechin, Floating-Point Datapaths in CIRCT via HArCo and harico-arith-to-comb Lowering, in *EuroLLVM Developers Meeting 2026*, Dublin, Ireland. • HAL:05576131.
- [Led+26a] **L. Ledoux**, P. Cochard, and F. de Dinechin, Arithmetic lowering with Emerald-MLIR: bridging tensor and DSP kernels to silicon datapaths, Aussois, France. • HAL:05489427.
- [Led+25a] **L. Ledoux**, P. Cochard, L. Forget, and F. de Dinechin, Towards multi-level arithmetic optimizations, in *EuroLLVM 2025*, Berlin, Germany. • HAL:05063466.
- [LC24b] **L. Ledoux** and M. Casas, LLM MMM: large language models matrixmatrix multiplications characterization on open silicon, in *11th BSC Severo Ochoa Doctoral Symposium 2024*, Barcelona, Spain. • HAL:04592229.
- [LC24a] **L. Ledoux** and M. Casas, The grafted superset approach: bridging Python to silicon with asynchronous compilation and beyond, in *4th Workshop on Open-Source Design Automation (OSDA 2024)*, hosted at DATE, Valencia, Spain. • HAL:04587458.
- [LC23b] **L. Ledoux** and M. Casas, Open-source GEMM hardware kernels generator: toward numerically-tailored computations, in *10th BSC Severo Ochoa Doctoral Symposium 2023*, Barcelona, Spain. • arXiv:2305.18328 • HAL:04094835.

Invited Talks & Seminars

- [LC25] **L. Ledoux** and P. Cochard, FloPoCo and MLIR: a multi-level compilation framework for many intents, Holigrail Seminar, Sorbonne University, Paris, France, September 18, 2025. • HAL:05412130.
- [Led24] **L. Ledoux**, The walls and the dark silicon era: an arithmetic perspective, seminar at Inria Rennes (Team TARAN), Rennes, France, December 2, 2024. • HAL:04823130.
- [LC19] **L. Ledoux** and M. Casas, Accelerating DL inference with (Open)CAPI and posit numbers, in *OpenPOWER Summit Europe 2019*, Lyon, France. • HAL:04094850.

Academic Reviewing

2026 FCCM (×1), ASAP (×2).

Conference Organization Experience

- LAC25 **Linux Audio Conference**, Lyon, France (2025). *Technician*. Responsible for sound, YouTube streaming, and microphone management. <https://jimlac25.inria.fr/lac/>
- JIM25 **Journées de l'Informatique Musicale**, Lyon, France (2025). *Technician*. Responsible for sound, YouTube streaming, and microphone management. <https://jimlac25.inria.fr/jim/>
- HC32 **HotChips 32nd**, Remote (Digital), SARS-CoV-2 Period (2020). *Remote Assistant*. Assisted with digital entrance management, conference link distribution, and attendee support during the event. <https://hc32.hotchips.org/>
- FPL29 **Field-Programmable Logic Conference 29th**, Barcelona, Spain (2019). *Reception and Entrance Coordinator*. Managed reception desk activities, including badge distribution and providing microphones for speakers during presentations. <https://fpl2019.bsc.es/>

Internship Mentoring

- 2026 – Now **Esteban Wybouw**, “Porting FloPoCo to HArCo: an IR-based core generation framework for multi-HDL hardware generation,” co-supervised with Florent de Dinechin and Pierre Cochard.

Academic Summer Schools and Workshops

- ACM21 **ACM Europe Summer School on High Performance Computing**, Barcelona, Spain (2021). *Participant*. Attended lectures and hands-on sessions covering exascale architectures, programming models, and numerical methods. <https://europe.acm.org/summer-school>
- YP19 **Yale Patts Computer Architecture Summer School**, Barcelona, Spain (2019). *Participant*. Attended the special summer edition of Prof. Yale Patts course on out-of-order execution and advanced microarchitectural design.
- XIL19 **Xilinx Machine Learning and Artificial Intelligence Seminar**, Paris, France (2019). *Participant*. Technical seminar focused on machine learning workloads and AI acceleration platforms.
- PRACE19 **PRACE Summer School on Heterogeneous Programming with OmpSs@FPGA**, Barcelona, Spain (2019). *Participant*. Training on FPGA-based acceleration and heterogeneous programming using the OmpSs@FPGA model.

Hardware & Software Realizations

- Tapeouts TinyTapeout GF0P2 (Faust→MLIR, tanh saturation), TinyTapeout Sky130 (modified placement flow), and MPW5 Sky130 (Posit/Quire systolic array). MPW1 and MPW4 contributions.
- Toolchains OSFNTC (numerically-tailored GEMM generator with PyTorch/OpenBLAS/OpenCAPI integration), SUF (Python→ASIC design-space exploration), VH2V (VHDL→Verilog for OpenLane/OpenROAD), POF (SystemVerilog Posit operators), gdsiistl (GDSII→STL).
- Open-Source Contributions CIRCT/LLVM (arith dialect + HDL lowering), oc-accel and capi2-bsp (OpenCAPI/CAPi2 support), OpenROAD global placer experiments/visualization, FloPoCo (Virtex UltraScale+ timing model).

Languages

- French Native
- Spanish Native (with an honest French accent)
- English Full Proficiency